

Synopsis - Deep Learning Mini Project

Title of the Project:

On-Device Acoustic Event Detection: A Comparative Study of CNN, CRNN, and Transformer Architectures under the Accuracy–Efficiency Trade-off

Team No. and Names:

Team No.: _____

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Brief Description of the Project:

Environmental sound classification — recognising events such as sirens, car horns, dog barks, drilling, and street music from short audio clips — is a core problem in acoustic scene analysis, with applications in smart cities, assistive devices, surveillance, and mobile sensing. In most of these applications the model must run on the edge device that captures the audio, where compute, memory, and battery are constrained. Classification accuracy alone is therefore an incomplete picture: a model that is marginally more accurate but far larger or slower may be unusable in deployment. This project studies the accuracy–efficiency trade-off directly.

We formulate the task as multi-class classification over the ten sound categories of the UrbanSound8K dataset. All audio is converted to a common front-end representation — log-mel spectrograms — so that every model consumes identical inputs under a shared, reproducible pipeline. On top of this representation we implement and compare three deep-learning architectures drawn from distinct families: (i) a convolutional neural network (CNN), including a mobile-native variant (MobileNetV3 / EfficientNet-Lite); (ii) a convolutional-recurrent network (CRNN) that pairs a CNN feature extractor with a bidirectional recurrent layer to model temporal structure; and (iii) an attention-based Audio Spectrogram Transformer (AST), fine-tuned from a pretrained checkpoint. A model-compression axis (post-training quantization and pruning) is applied to characterise how each architecture behaves when reduced toward edge-deployable size.

All models are trained and evaluated under a single common protocol: the dataset's predefined 10-fold cross-validation split, identical preprocessing, and a shared metric set. We report classification accuracy and macro-averaged F1 for predictive quality, and parameter count, on-disk model size, and inference latency (CPU, with an optional on-device mobile measurement) for efficiency. Results are consolidated into a single comparison table spanning all models and metrics, accompanied by an error analysis of which sound classes each model confuses and a discussion of why performance and efficiency differ across architecture families.

The intended contribution of the mini-project is an evidence-based comparison of how three representative deep-learning families trade predictive performance against deployability on a standard acoustic benchmark, rather than a pursuit of state-of-the-art accuracy alone. Each of the three architecture families is owned end-to-end by one team member, so that every member independently implements, tunes, evaluates, and defends a distinct model.

Datasets Identified:

UrbanSound8K — a widely used benchmark for urban/environmental sound classification.

- Content: 8,732 labelled audio excerpts (≤ 4 seconds each) of urban sounds.
- Classes (10): air conditioner, car horn, children playing, dog bark, drilling, engine idling, gun shot, jackhammer, siren, street music.
- Structure: audio provided as WAV files, pre-sorted into 10 stratified folds for cross-validation, with accompanying metadata (class label, fold, salience, source ID).
- Evaluation protocol: the official 10-fold split is used for train/validation/test so results are directly comparable across all models and reproducible.
- Front-end: audio is resampled to a fixed rate and converted to log-mel spectrograms as the common model input.
- Availability: publicly available for research use from the dataset authors (Salamon et al., NYU).

Proposed Models for Comparison:

- Model 1 — CNN (convolutional family): a spectrogram-based CNN baseline plus a mobile-native variant (MobileNetV3 / EfficientNet-Lite) representing the 'efficient by design' point.
- Model 2 — CRNN (convolutional-recurrent family): a CNN feature extractor followed by a bidirectional LSTM/GRU to capture temporal dynamics.
- Model 3 — Audio Spectrogram Transformer (attention family): a pretrained AST fine-tuned on the dataset, with a quantized/pruned variant for the efficiency comparison.

A model-compression axis (post-training quantization, pruning) is applied across the models to characterise the accuracy–efficiency frontier, providing the fourth comparison dimension required for a meaningful study.

Model-to-Member Assignment:

- Rohan Sinha (230911005) — CNN baseline + mobile-native variant; owns the shared spectrogram preprocessing pipeline and the efficiency-benchmarking harness (params, size, latency).
- Akshdeep Singh (230911466) — CRNN (CNN + BiLSTM/GRU); implementation, tuning, and temporal-modelling evaluation.

- Siddharth Nair (230953284) — Audio Spectrogram Transformer (fine-tuned) + its compressed variant; attention-model implementation and quantization/pruning study.

Existing State-of-the-Art Methods:

Deep CNNs on log-mel spectrograms are the established baseline for environmental sound classification (e.g., PiczakCNN, SB-CNN with data augmentation on UrbanSound8K). CRNN architectures are widely used in sound event detection to combine convolutional feature extraction with recurrent temporal modelling. More recently, the Audio Spectrogram Transformer (AST) applies a pure-attention architecture to audio and achieves strong results on AudioSet/ESC-50. On the efficiency side, mobile-native backbones (MobileNet, EfficientNet-Lite) and model-compression techniques (quantization, pruning, knowledge distillation) are the standard tools for edge deployment. This project situates its comparison against these methods.

Objectives:

- To implement and compare three deep-learning architecture families (CNN, CRNN, Transformer) for environmental sound classification on UrbanSound8K under a single common evaluation protocol.
- To characterise the accuracy–efficiency trade-off of these architectures by jointly measuring predictive performance (accuracy, macro-F1) and deployability (parameters, model size, inference latency), including under model compression.
- To identify which architecture family offers the most favourable accuracy-per-cost point for on-device acoustic event detection, supported by a comparative results table and error analysis.

Importance / Motivation:

Acoustic event detection increasingly runs on resource-constrained edge devices — phones, wearables, and embedded sensors — where the audio is captured. In such settings the practically relevant question is not merely which model is most accurate, but which delivers acceptable accuracy within tight compute, memory, and latency budgets. Much prior work reports accuracy in isolation; comparatively less systematically characterises how different architecture families trade accuracy against efficiency on a common benchmark. By evaluating CNN, CRNN, and Transformer models under one protocol and one comparison table that includes size and latency alongside accuracy, this project produces directly actionable guidance for deploying acoustic classifiers on-device, and gives each team member independent ownership of a distinct, viva-defensible model.

Preliminary Reference List:

- [1] Salamon, J., Jacoby, C., & Bello, J. P. (2014). A Dataset and Taxonomy for Urban Sound Research. In Proceedings of the 22nd ACM International Conference on Multimedia (ACM-MM), 1041–1044.
- [2] Salamon, J., & Bello, J. P. (2017). Deep Convolutional Neural Networks and Data Augmentation for Environmental Sound Classification. IEEE Signal Processing Letters, 24(3), 279–283.
- [3] Piczak, K. J. (2015). Environmental Sound Classification with Convolutional Neural Networks. In IEEE International Workshop on Machine Learning for Signal Processing (MLSP), 1–6.
- [4] Gong, Y., Chung, Y.-A., & Glass, J. (2021). AST: Audio Spectrogram Transformer. In Proceedings of Interspeech 2021, 571–575.
- [5] Cakır, E., Parascandolo, G., Heittola, T., Huttunen, H., & Virtanen, T. (2017). Convolutional Recurrent Neural Networks for Polyphonic Sound Event Detection. IEEE/ACM Transactions on Audio, Speech, and Language Processing, 25(6), 1291–1303.
- [6] Howard, A., Sandler, M., Chu, G., et al. (2019). Searching for MobileNetV3. In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 1314–1324.
- [7] Jacob, B., Kligys, S., Chen, B., et al. (2018). Quantization and Training of Neural Networks for Efficient Integer-Arithmetic-Only Inference. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2704–2713.

Work Plan:

Sl. No.	Component / Work Element / Milestone	Expected Completion (Day/Month)
1	Dataset acquisition, exploratory analysis, and log-mel spectrogram preprocessing pipeline (shared)	05 Sep 2026
2	CNN baseline + mobile-native variant (MobileNetV3/EfficientNet-Lite) — implementation & preliminary results	18 Sep 2026
3	CRNN (CNN + BiLSTM/GRU) — implementation & preliminary results	22 Sep 2026
4	Interim submission: literature table (8–10 papers), preprocessing + ≥2 models with results, contribution log	25 Sep 2026
5	Audio Transformer (AST, fine-tuned) — implementation & tuning	12 Oct 2026
6	Model compression axis (quantization/pruning) + latency & size benchmarking across all models	20 Oct 2026
7	Comparative evaluation, error analysis, comparison table, final report & presentation	29 Oct 2026